

Machine Learning for Biostatistics

Module 6

Armando Teixeira-Pinto & Jaroslaw Harezlak

2026-09-04

Contents

Beyond Additivity	5
Introduction	5
Dataset used in the examples	5
Slides from the videos	6
1 Regression and Classification Trees	7
1.1 Introduction	7
1.2 Readings	10
1.3 Practice session	10
1.4 Exercises	26
2 Bagging	27
2.1 Introduction	27
2.2 Readings	28
2.3 Practice session	28
2.4 Exercises	33
3 Random Forests	35
3.1 Introduction	35
3.2 Readings	36
3.3 Practice session	36
3.4 Exercises	41

Beyond Additivity

Introduction

So far, most of the methods that we have seen (with the exception of KNN) assume an additive effect of the predictors. We will now study non-parametric methods to estimate $f(\mathbf{x})$.

By the end of this module you should be able to:

1. Construct regression and classification trees
2. Use cross-validation to prune a tree
3. Understand the advantages and disadvantages of trees
4. Define bagging and boosting
5. Fit random forests

Dataset used in the examples

The dataset **triceps** is available in the **MultiKink** package. You may `install.packages("MultiKink")`, load the library (`library(MultiKink)`) and then run `data("triceps")`.

The data are derived from an anthropometric study of 892 females under 50 years in three Gambian villages in West Africa. There are 892 observations on the following 3 variables:

- age - Age of respondents.
- lntriceps - Log of the triceps skinfold thickness.
- triceps - Triceps skinfold thickness.

The data `SA_heart.csv` is retrospective sample of males in a heart-disease high-risk region of the Western Cape, South Africa. There are roughly two controls per case of CHD.

Many of the CHD positive men have undergone blood pressure reduction treatment and other programs to reduce their risk factors after their CHD event. In

some cases the measurements were made after these treatments. These data are taken from a larger dataset, described in Rousseau et al, 1983, South African Medical Journal.

The data contains 462 observations on the following 10 variables.

- sbp - systolic blood pressure
- tobacco - cumulative tobacco (kg)
- ldl - low density lipoprotein cholesterol
- adiposity - a numeric vector
- famhist - family history of heart disease, a factor with levels Absent Present
- typea - type-A behavior
- obesity - a numeric vector
- alcohol - current alcohol consumption
- age - age at onset
- chd- response, coronary heart disease (1 - chd, 0 - no chd)

The file SBI.csv contains 2349 records of children admitted to the emergency room with fever and tested for serious bacterial infection (**sbi**). The following variables are included :

- id – patient’s number
 - fever_hours – duration of the fever in hours
 - age – child’s age
 - sex – child’s sex (M / F)
 - wcc – white cell count
 - prevAB – previous antibiotics (Yes / No)
 - sbi – serious bacterial infection (Not Applicable / UTI / Pneum / Bact)
 - pct – procalcitonin
 - crp – c-reactive protein
-

Slides from the videos

You can download the slides used in the videos from Beyond Additivity:

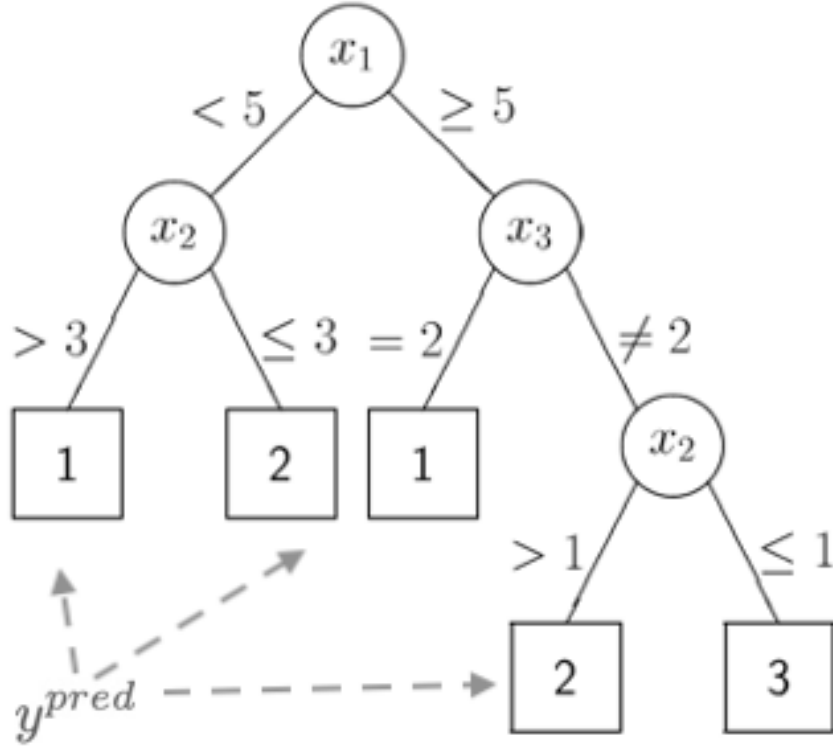
Slides

Chapter 1

Regression and Classification Trees

1.1 Introduction

Decision trees can be used for continuous outcomes - *regression trees* - or categorical ones - *classification trees*. Classification and regression trees are sometimes referred as **CART**.



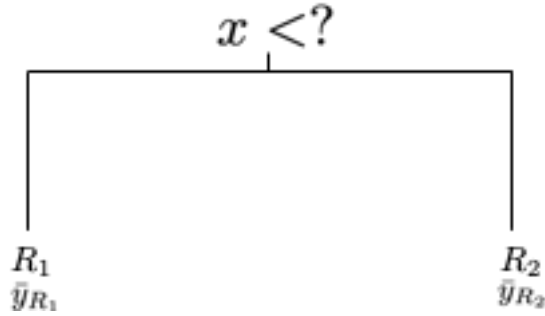
The figure above represents a classification tree with several branches that partition the predictors space into disjoint regions according to some loss function. The predicted value for the outcome is found at the end of the branches.

For **regression trees**, and similarly to the linear model, the predicted value will correspond to the mean of the region created by the partition. For a classification tree, the prediction will correspond to a category of the outcome.

In regression trees, a common loss function used to generate the partition is:

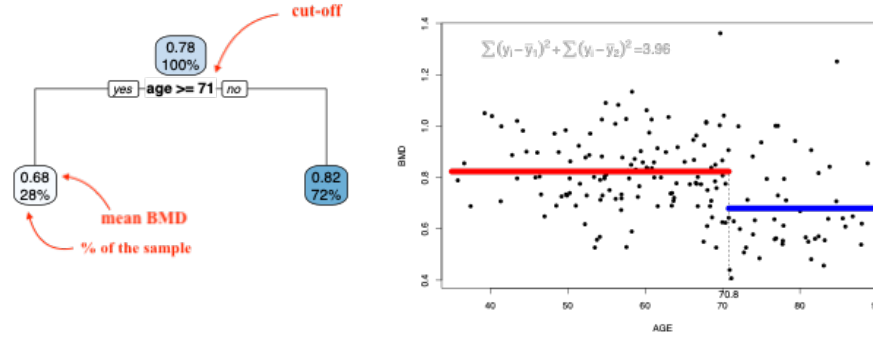
$$\sum_{i \in R_1} (y_i - \bar{y}_{R_1})^2 + \sum_{i \in R_2} (y_i - \bar{y}_{R_2})^2$$

where $\bar{y}_{R_1}$ and $\bar{y}_{R_2}$ are the means of the outcome for the regions created by splitting the region defined by a predictor into 2 parts.



The animation below shows different splits of the predictor and the correspondent prediction (mean of y) - the two horizontal lines represent to the mean of the outcome in the two regions. The change in the loss function is computed in the top left corner:

At each step, the predictor and respective cutoff that minimise the loss function is selected to split the branch.



We can keep dividing the branches until each region has 1 data point (there can be more if there are ties). This of course will result in the tree overfitting the data. We would prefer a smaller tree that would result in a lower MSE. Cutting the branches of the tree to reduce its complexity and potentially leading to an improvement in the MSE is called *pruning*.

Once again we will use the idea of penalisation to balance complexity and fitting.

$$\sum_{m=1}^{|T|} \sum_{x_i \in R_m} (y_i - \bar{y}_{R_m})^2 + \alpha |T|, |T| \text{ is the number of nodes in the tree}$$

The penalisation (tuning) parameter, α , controls the amount of pruning and it can be chosen by cross-validation.

Classification trees follow a similar logic but the outcome is categorical. We will have to consider a different loss function. Two common loss functions used

in classification trees are:

Gini index: $1 - \sum_{k=1}^K p_k^2$

- Favours larger partitions.
- Perfectly classified, Gini Index would be zero.
- Evenly distributed would be $1 - (1/\# \text{ Classes})$.

Information gain (entropy): $\sum_{k=1}^K -p_k \log_2 p_k$

- Favours splits with small counts but many unique values
- Weights probability of class by $\log_2$ of the class probability
- Information Gain is the entropy of the parent node minus the entropy of the child nodes.

1.2 Readings

Read the following chapters of *An introduction to statistical learning*:

- 8.1 The Basics of Decisions Trees
- An Introduction to Recursive Partitioning Using the RPART Routines - extra notes

1.3 Practice session

Task 1 - Grow a complete regression tree

Using the BMD.csv dataset,

Using the bmd.csv, fit a tree to predict bone mineral density (BMD) based on AGE

```
#libraries that we will need
set.seed(1974) #fix the random generator seed

library(rpart) #library for CART
library(rpart.plot)
#read the dataset
bmd.data <-
  read.csv("https://www.dropbox.com/s/c6mhgatkotuze8o/bmd.csv?dl=1",
           stringsAsFactors = TRUE)
```

The package `rpart` implements the classification and regression trees and the `rpart.plot` contains the tree plotting function. We will use the `rpart()` function to fit the tree, with some options to grow the complete tree

```
library(caret)
t1 <- rpart(bmd ~ age,
            data = bmd.data,
```

```

method = "anova",          #indicates the outcome is continuous
control = rpart.control(
  minsplit = 1, # min number of observ for a split
  minbucket = 1, # min nr of obs in terminal nodes
  cp=0)           #decrease in complex for a split
)

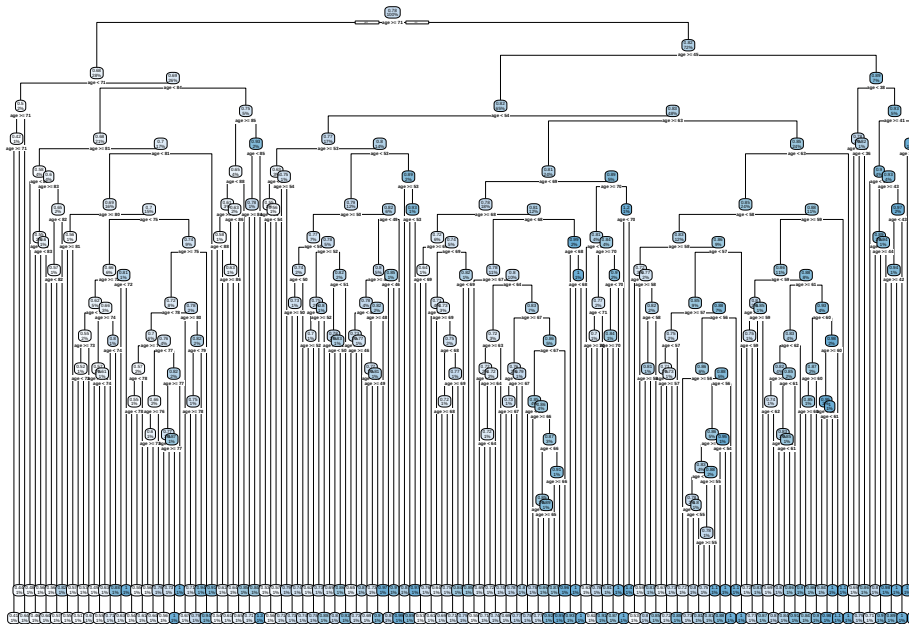
```

```

#the rpart.plot() may take a long time to plot
#the complete tree
#if you cannot run it, just try plot(t1); text(t1, pretty=1)
#and you will see just the structure of the tree
rpart.plot(t1)

```

Warning: labs do not fit even at cex 0.15, there may be some overplotting



We can now prune the tree using a limit for the complexity parameter cp . This will indicate that only a split with CP higher than the limit is worth it.

```

printcp(t1)           #CP for the complete tree

```

```

##
## Regression tree:
## rpart(formula = bmd ~ age, data = bmd.data, method = "anova",
##       control = rpart.control(minsplit = 1, minbucket = 1, cp = 0))
##
## Variables actually used in tree construction:

```

```

## [1] age
##
## Root node error: 4.659/169 = 0.027568
##
## n= 169
##
##          CP nsplit  rel error  xerror   xstd
## 1  1.5011e-01      0 1.0000e+00 1.01765 0.11147
## 2  2.2857e-02      1 8.4989e-01 0.89504 0.11530
## 3  2.1625e-02      2 8.2703e-01 1.18928 0.15101
## 4  2.0621e-02      8 6.9728e-01 1.21668 0.15171
## 5  1.7896e-02     11 6.2374e-01 1.29781 0.16189
## 6  1.6691e-02     13 5.8795e-01 1.29128 0.16179
## 7  1.3855e-02     14 5.7126e-01 1.27044 0.15896
## 8  1.2133e-02     15 5.5740e-01 1.28841 0.15934
## 9  1.1410e-02     16 5.4527e-01 1.35628 0.16661
## 10 1.0975e-02     17 5.3386e-01 1.40868 0.17009
## 11 1.0843e-02     21 4.8996e-01 1.40868 0.17009
## 12 1.0550e-02     26 4.3574e-01 1.42386 0.17023
## 13 1.0318e-02     27 4.2519e-01 1.39577 0.16607
## 14 9.5921e-03     28 4.1488e-01 1.43436 0.17090
## 15 9.5636e-03     29 4.0528e-01 1.43302 0.17121
## 16 9.4801e-03     35 3.4511e-01 1.43302 0.17121
## 17 8.8498e-03     37 3.2615e-01 1.45697 0.17171
## 18 8.7833e-03     38 3.1730e-01 1.46385 0.17148
## 19 8.6510e-03     39 3.0851e-01 1.46385 0.17148
## 20 8.1753e-03     42 2.8164e-01 1.51318 0.17653
## 21 8.1257e-03     43 2.7347e-01 1.50180 0.17666
## 22 7.9341e-03     45 2.5722e-01 1.49873 0.17666
## 23 6.8660e-03     46 2.4928e-01 1.48046 0.17148
## 24 6.2708e-03     47 2.4242e-01 1.52337 0.17402
## 25 5.9903e-03     52 2.1106e-01 1.51988 0.16810
## 26 5.9250e-03     54 1.9908e-01 1.53343 0.16805
## 27 5.9229e-03     56 1.8723e-01 1.53729 0.16825
## 28 5.8272e-03     57 1.8131e-01 1.53729 0.16825
## 29 5.4523e-03     58 1.7548e-01 1.54174 0.16837
## 30 5.0336e-03     59 1.7003e-01 1.57444 0.16855
## 31 4.9113e-03     60 1.6500e-01 1.57002 0.16862
## 32 4.6286e-03     63 1.5026e-01 1.56988 0.17224
## 33 4.5364e-03     64 1.4563e-01 1.62250 0.17986
## 34 4.3127e-03     66 1.3656e-01 1.62250 0.17986
## 35 4.3104e-03     68 1.2794e-01 1.62250 0.17986
## 36 4.2807e-03     69 1.2363e-01 1.62704 0.17972
## 37 3.9800e-03     71 1.1506e-01 1.60725 0.17771
## 38 3.6958e-03     72 1.1108e-01 1.62131 0.17727
## 39 3.1052e-03     75 9.9997e-02 1.61935 0.17755

```

## 40	3.0222e-03	76	9.6892e-02	1.62050	0.17739
## 41	2.7468e-03	78	9.0848e-02	1.61019	0.17499
## 42	2.6775e-03	79	8.8101e-02	1.61092	0.17494
## 43	2.5828e-03	81	8.2746e-02	1.62495	0.17811
## 44	2.5347e-03	82	8.0163e-02	1.62522	0.17810
## 45	2.4665e-03	83	7.7628e-02	1.60648	0.17644
## 46	2.4609e-03	84	7.5162e-02	1.61453	0.17668
## 47	2.4243e-03	86	7.0240e-02	1.61426	0.17669
## 48	2.4140e-03	87	6.7816e-02	1.61381	0.17671
## 49	2.3335e-03	88	6.5402e-02	1.61381	0.17671
## 50	2.3321e-03	89	6.3068e-02	1.62015	0.17692
## 51	2.2471e-03	91	5.8404e-02	1.62341	0.17707
## 52	2.1072e-03	92	5.6157e-02	1.62288	0.17563
## 53	2.0870e-03	93	5.4050e-02	1.64493	0.17924
## 54	2.0451e-03	100	3.7574e-02	1.64696	0.17917
## 55	1.9386e-03	101	3.5529e-02	1.64164	0.17923
## 56	1.8005e-03	102	3.3590e-02	1.64927	0.17891
## 57	1.3686e-03	103	3.1790e-02	1.65874	0.17933
## 58	1.3438e-03	107	2.6306e-02	1.66822	0.17937
## 59	1.2138e-03	108	2.4962e-02	1.66544	0.17949
## 60	1.1334e-03	110	2.2535e-02	1.65286	0.17945
## 61	1.0723e-03	111	2.1401e-02	1.66121	0.17957
## 62	1.0605e-03	113	1.9257e-02	1.66661	0.17950
## 63	1.0035e-03	114	1.8196e-02	1.67577	0.18096
## 64	9.4464e-04	115	1.7193e-02	1.68613	0.18094
## 65	7.5924e-04	116	1.6248e-02	1.69531	0.18176
## 66	7.5061e-04	118	1.4730e-02	1.70108	0.18176
## 67	7.4468e-04	119	1.3979e-02	1.70117	0.18175
## 68	7.3755e-04	120	1.3234e-02	1.70117	0.18175
## 69	7.1811e-04	121	1.2497e-02	1.70013	0.18179
## 70	7.1728e-04	122	1.1779e-02	1.70013	0.18179
## 71	7.1635e-04	123	1.1061e-02	1.70013	0.18179
## 72	7.1110e-04	124	1.0345e-02	1.70013	0.18179
## 73	6.6133e-04	125	9.6340e-03	1.70245	0.18178
## 74	6.5629e-04	126	8.9727e-03	1.69886	0.18200
## 75	6.5461e-04	127	8.3164e-03	1.69886	0.18200
## 76	5.8428e-04	128	7.6618e-03	1.71271	0.18347
## 77	5.1243e-04	129	7.0775e-03	1.71597	0.18319
## 78	4.7746e-04	130	6.5651e-03	1.71423	0.18334
## 79	4.7460e-04	131	6.0876e-03	1.71405	0.18335
## 80	4.5064e-04	132	5.6130e-03	1.71288	0.18339
## 81	4.3443e-04	133	5.1624e-03	1.71219	0.18323
## 82	4.0723e-04	134	4.7279e-03	1.71290	0.18328
## 83	4.0723e-04	135	4.3207e-03	1.71573	0.18329
## 84	4.0195e-04	136	3.9135e-03	1.71573	0.18329
## 85	3.6728e-04	137	3.5115e-03	1.71298	0.18333

```

## 86 3.3896e-04 138 3.1442e-03 1.71664 0.18344
## 87 3.1877e-04 139 2.8053e-03 1.71664 0.18344
## 88 3.1811e-04 140 2.4865e-03 1.71680 0.18344
## 89 2.7369e-04 141 2.1684e-03 1.71482 0.18318
## 90 2.0990e-04 142 1.8947e-03 1.71456 0.18320
## 91 1.8394e-04 143 1.6848e-03 1.71441 0.18311
## 92 1.8306e-04 144 1.5008e-03 1.71669 0.18317
## 93 1.8237e-04 145 1.3178e-03 1.71669 0.18317
## 94 1.3863e-04 146 1.1354e-03 1.71682 0.18316
## 95 1.1758e-04 148 8.5816e-04 1.71809 0.18298
## 96 9.2318e-05 149 7.4058e-04 1.71952 0.18324
## 97 9.1614e-05 150 6.4826e-04 1.71872 0.18323
## 98 8.9016e-05 152 4.6503e-04 1.71872 0.18323
## 99 8.1752e-05 153 3.7601e-04 1.71872 0.18323
## 100 5.6280e-05 154 2.9426e-04 1.71817 0.18325
## 101 5.0071e-05 155 2.3798e-04 1.72019 0.18319
## 102 4.1650e-05 156 1.8791e-04 1.71880 0.18322
## 103 2.7818e-05 157 1.4626e-04 1.71833 0.18317
## 104 2.2876e-05 158 1.1844e-04 1.71756 0.18313
## 105 1.7583e-05 159 9.5567e-05 1.71683 0.18313
## 106 1.5454e-05 160 7.7983e-05 1.71756 0.18315
## 107 1.3947e-05 161 6.2529e-05 1.71734 0.18318
## 108 1.3462e-05 162 4.8582e-05 1.71734 0.18318
## 109 1.2059e-05 163 3.5120e-05 1.71677 0.18317
## 110 1.2058e-05 164 2.3061e-05 1.71677 0.18317
## 111 6.8126e-06 165 1.1003e-05 1.71677 0.18317
## 112 2.4728e-06 166 4.1899e-06 1.71691 0.18318
## 113 1.7171e-06 167 1.7171e-06 1.71682 0.18318
## 114 0.0000e+00 168 0.0000e+00 1.71656 0.18312

prune.t1 <- prune(t1, cp=0.02) #prune the tree with cp=0.02

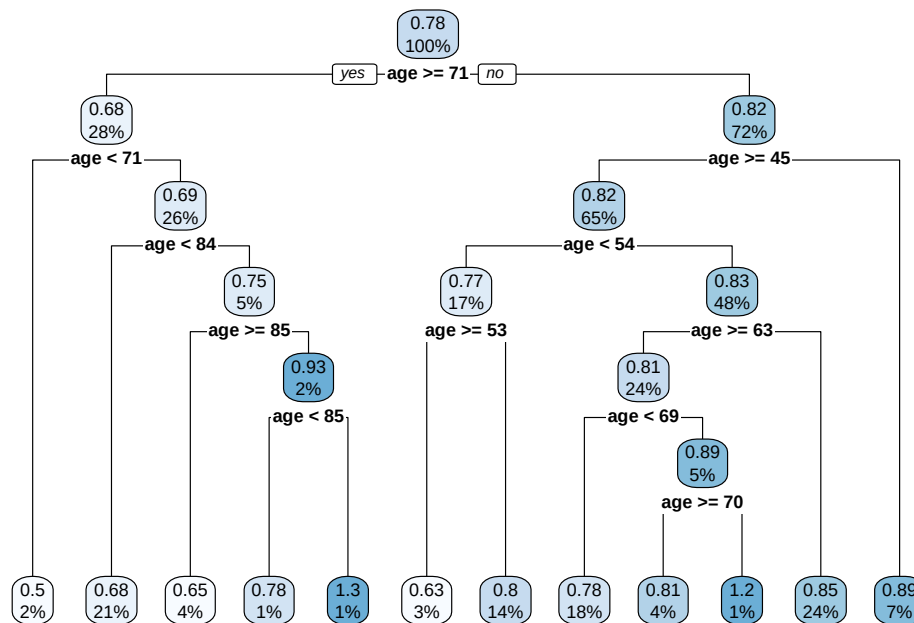
printcp(prune.t1)

##
## Regression tree:
## rpart(formula = bmd ~ age, data = bmd.data, method = "anova",
##       control = rpart.control(minsplit = 1, minbucket = 1, cp = 0))
##
## Variables actually used in tree construction:
## [1] age
##
## Root node error: 4.659/169 = 0.027568
##
## n= 169
##

```

```
##          CP nsplit rel error  xerror   xstd
## 1 0.150111      0   1.00000 1.01765 0.11147
## 2 0.022857      1   0.84989 0.89504 0.11530
## 3 0.021625      2   0.82703 1.18928 0.15101
## 4 0.020621      8   0.69728 1.21668 0.15171
## 5 0.020000     11   0.62374 1.29781 0.16189
```

```
rpart.plot(prune.t1) #pruned tree
```

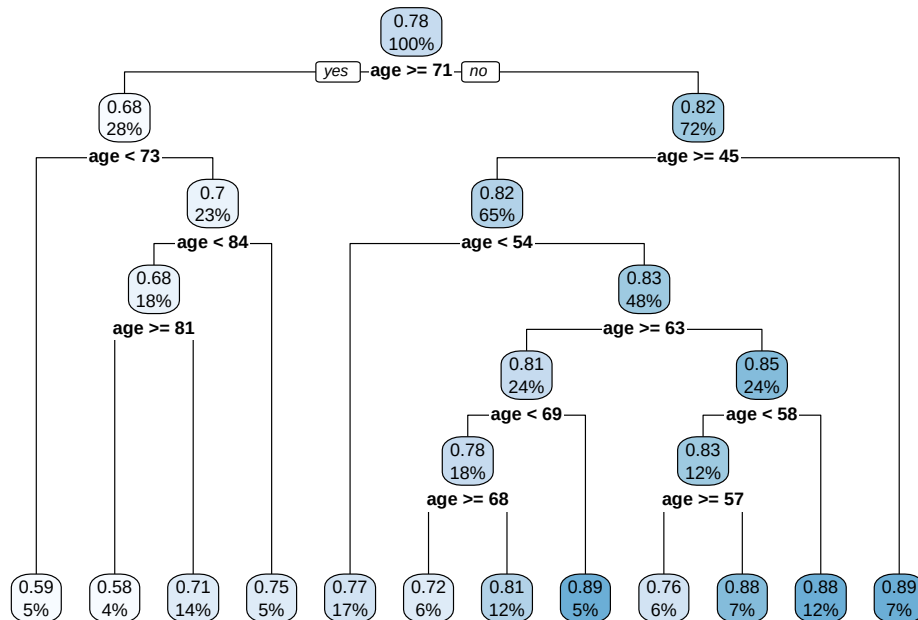


TRY IT YOURSELF:

- 1) Get a tree for the problem above but with the defaults of `rpart.control()`.

See the solution code

```
?rpart.control #see the defaults
t1.default <- rpart(bmd ~ age,
  data = bmd.data,
  method = "anova")
rpart.plot(t1.default)
```



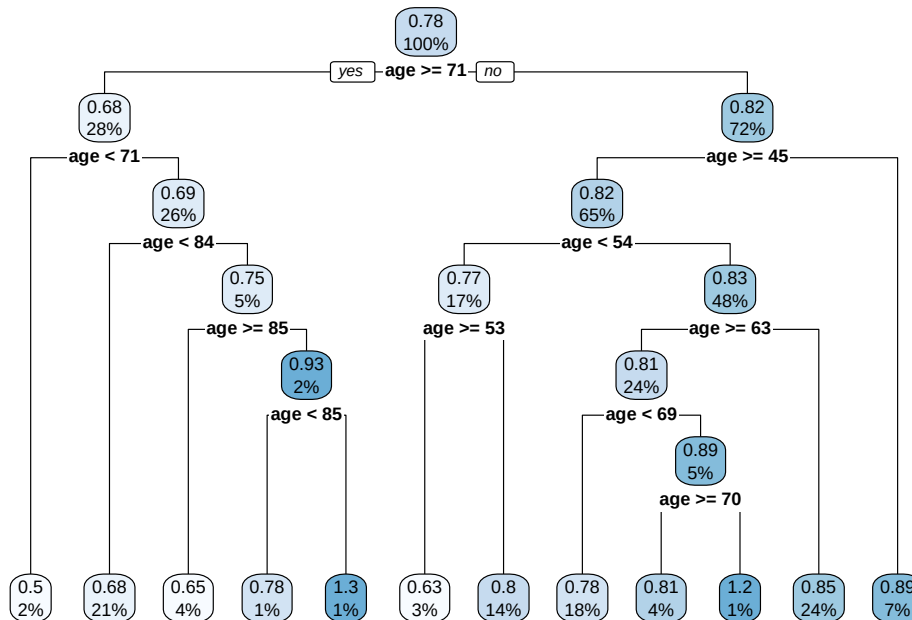
2) Use the `caret` package to fit the `prune.t1` tree

See the solution code

```
trctrl <- trainControl(method = "cv", number = 10)

#to get the same tree we will fix the tuning parameter cp
#to be 0.02
t1.caret <- train(bmd ~ age,
                  data = bmd.data,
                  method = "rpart",
                  trControl=trctrl,
                  control = rpart.control(minsplit = 1,
                                          minbucket = 1,
                                          cp=0.02),
                  tuneGrid = expand.grid(cp=0.02))

rpart.plot(t1.caret$finalModel)
```

Task 2 - Make predictions from a tree

Let's use the pruned tree from task 1 (`pruned.t1`) to predict the BMD for an individual 70 years old and compare it with the predictions from a linear model

```
lm1 <- lm(bmd ~ age,
          data = bmd.data)

predict(prune.t1,
        newdata = data.frame(age=70))      #prediction using the tree

predict(lm1,
        newdata = data.frame(age=70))      #prediction using the linear model
```

We now want to plot the predictions from the tree for ages 40 through 90 together with the predictions from the linear model

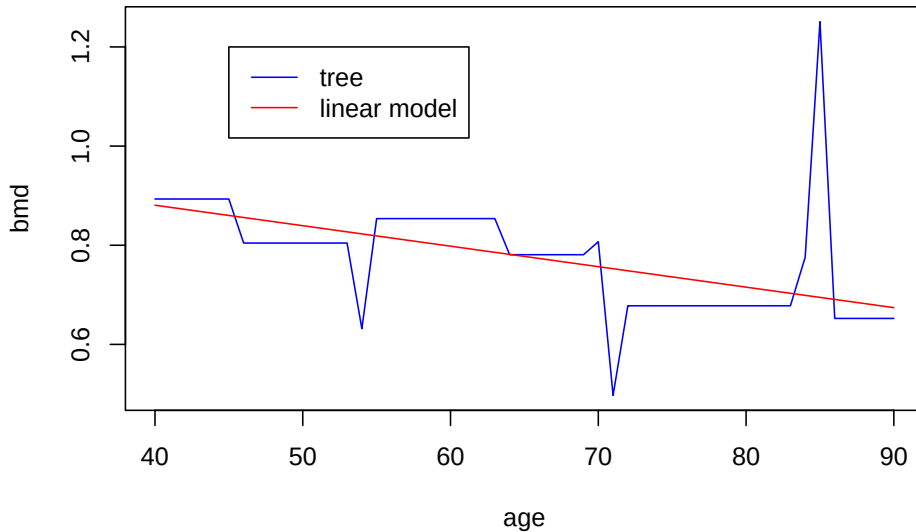
```
pred.tree <- predict(prune.t1,
                    newdata = data.frame(age=seq(40,90,1)))

pred.lm <- predict(lm1,
                  newdata = data.frame(age=seq(40,90,1)))

plot(seq(40,90,1), pred.tree,
     type = "l", col="blue",
     xlab="age", ylab="bmd")
```

```
lines(seq(40,90,1), pred.lm, col="red")

legend(45, 1.2, lty=c(1,1),
      c("tree", "linear model"),
      col=c("blue", "red"))
```



Task 3 - Fit a tree using Cross-validation

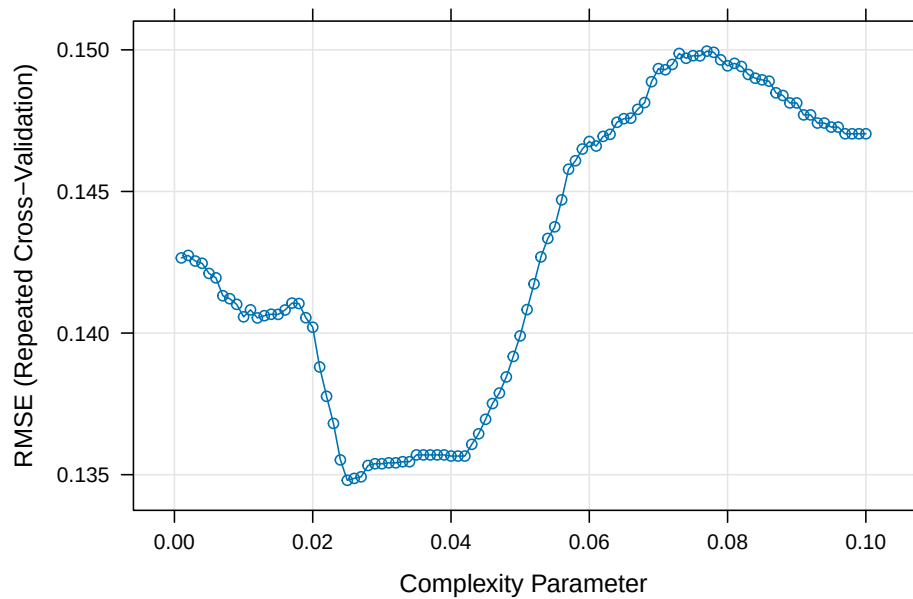
Let's fit now a tree to predict bone mineral density (BMD) based on AGE, SEX and BMI (BMI has to be computed) and compute the MSE.

We will use the `caret` package with `method="rpart"`. Notice that the `caret` will prune the tree based on the cross-validated `cp`.

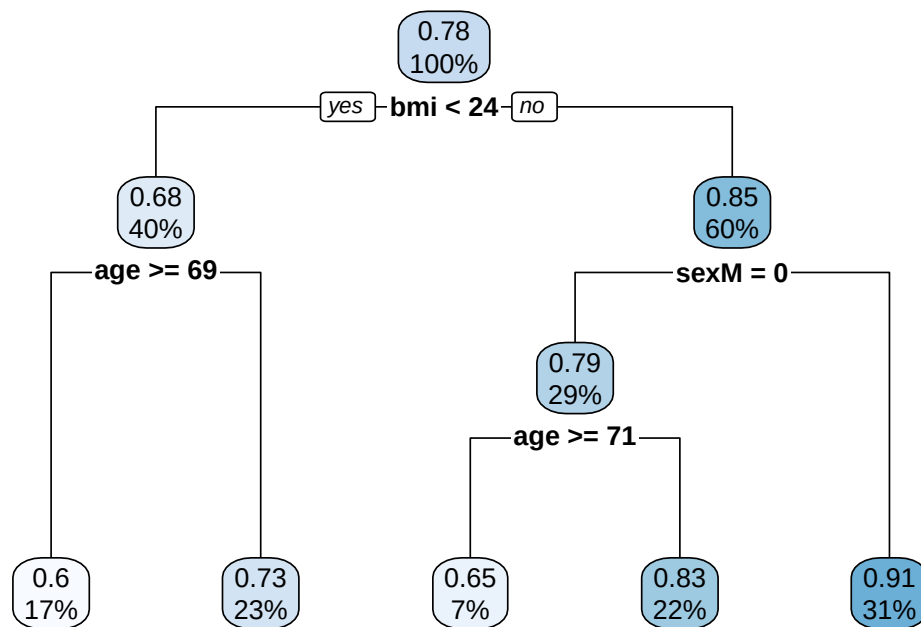
```
#compute BMI
bmd.data$bmi <- bmd.data$weight_kg / (bmd.data$height_cm/100)^2

trctrl <- trainControl(method = "repeatedcv", number = 10, repeats = 10)
t2.caret <- train(bmd ~ age + sex + bmi,
                  data = bmd.data,
                  method = "rpart",
                  trControl=trctrl,
                  tuneGrid = expand.grid(cp=seq(0.001, 0.1, 0.001))
                  )

#Plot the RMSE versus the CP
plot(t2.caret)
```



```
#Plot the tree with the optimal CP
rpart.plot(t2.caret$finalModel)
```



```
t2.caret$finalModel$cptable
```

```
##           CP nsplit rel error
## 1 0.26960125      0 1.0000000
```

```
## 2 0.07577601      1 0.7303988
## 3 0.06448418      2 0.6546227
## 4 0.05400352      3 0.5901386
## 5 0.02500000      4 0.5361350
```

We can compare the RMSE and R2 of the tree above with the linear model

```
trctrl <- trainControl(method = "repeatedcv", number = 5, repeats = 10)
lm2.caret <- train(bmd ~ age + sex + bmi,
                  data = bmd.data,
                  method = "lm",
                  trControl=trctrl
                  )

lm2.caret$results
```

```
##   intercept      RMSE Rsquared      MAE      RMSESD RsquaredSD      MAESD
## 1      TRUE 0.1383368 0.3292804 0.1046154 0.01878148 0.1145405 0.01176567
```

```
#extracts the row with the RMSE and R2 from the table of results
#corresponding to the cp with lowest RMSE (best tune)
t2.caret$results[t2.caret$results$cp==t2.caret$bestTune[1,1], ]
```

```
##      cp      RMSE Rsquared      MAE      RMSESD RsquaredSD      MAESD
## 25 0.025 0.134802 0.3819951 0.104164 0.02665188 0.1769088 0.01977811
```

Task 4 - Fit a classification tree

The SBI.csv dataset contains the information of more than 2300 children that attended the emergency services with fever and were tested for serious bacterial infection. The variable `sbi` has 4 categories: Not Applicable(no infection) / UTI / Pneum / Bact

Create a new variable `sbi.bin` that identifies if a child was diagnosed or not with serious bacterial infection.

```
set.seed(1999)
sbi.data <- read.csv("https://www.dropbox.com/s/wg32uj43fsy9yvd/SBI.csv?dl=1")
summary(sbi.data)
```

```
##      X      id      fever_hours      age      sex
## Min.   : 1.0   Min.   : 495   Min.   : 0.00   Min.   :0.010   Length :2348
## 1st Qu.:587.8  1st Qu.:133039  1st Qu.: 24.00  1st Qu.:0.760   N.unique : 2
## Median :1174.5 Median :160016  Median : 48.00  Median :1.525   N.blank  : 0
## Mean   :1174.5 Mean   :153698  Mean   : 80.06  Mean   :1.836   Min.nchar: 1
## 3rd Qu.:1761.2 3rd Qu.:196030  3rd Qu.: 78.00  3rd Qu.:2.752   Max.nchar: 1
## Max.   :2348.0 Max.   :229986  Max.   :3360.00 Max.   :4.990
##      sbi      pct      crp
## Length :2348 Min.   :8.647e-03 Min.   : 0.00
```

```
## N.unique :    4    1st Qu.:1.600e-01    1st Qu.: 11.83
## N.blank  :    0    Median :7.600e-01    Median : 30.97
## Min.nchar:    3    Mean   :3.744e+00    Mean   : 48.41
## Max.nchar:   13    3rd Qu.:4.620e+00    3rd Qu.: 66.20
##                Max.   :1.565e+02    Max.   :429.90

# Create a binary variable based on "sbi"
sbi.data$sbi.bin <- as.factor(ifelse(sbi.data$sbi == "NotApplicable", "NOSBI", "SBI"))
table(sbi.data$sbi, sbi.data$sbi.bin)

##
##                NOSBI  SBI
## Bact                0   34
## NotApplicable    1752    0
## Pneu                0  251
## UTI                0  311

#sbi.data.sub <- sbi.data[,c("sbi.bin", "fever_hours", "age", "sex", "wcc", "prevAB", "pct", "crp")]

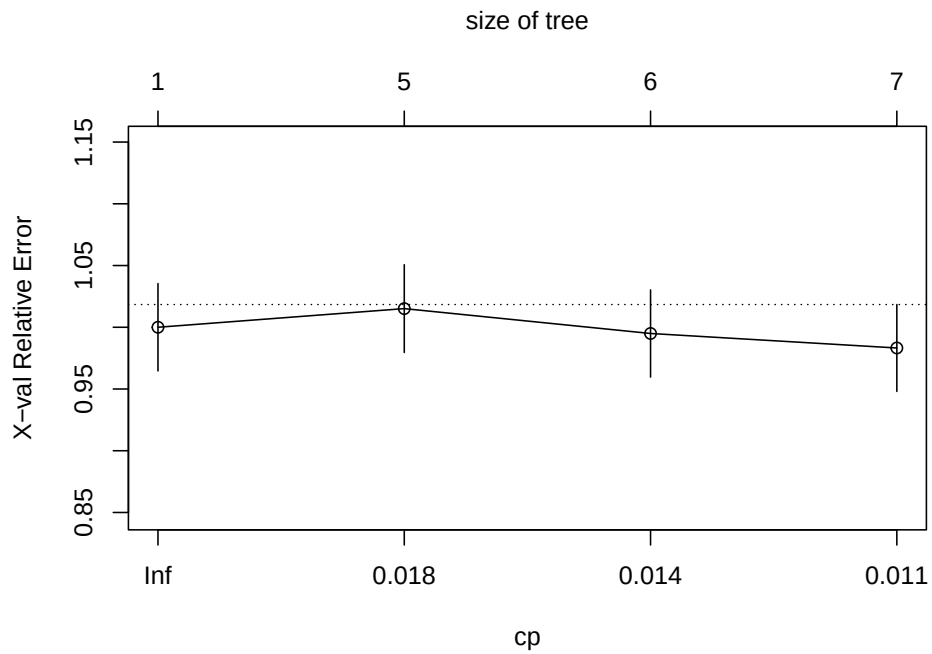
```

Now, build a classification tree to predict if a child has serious bacterial infection using **fever_hours**, **wcc**, **age**, **prevAB**, **pct**, and **crp**

```
sbi.tree <- rpart(sbi.bin ~ fever_hours+age+sex+wcc+prevAB+pct+crp,
                  data = sbi.data,
                  method = "class"
                  )

plotcp(sbi.tree)

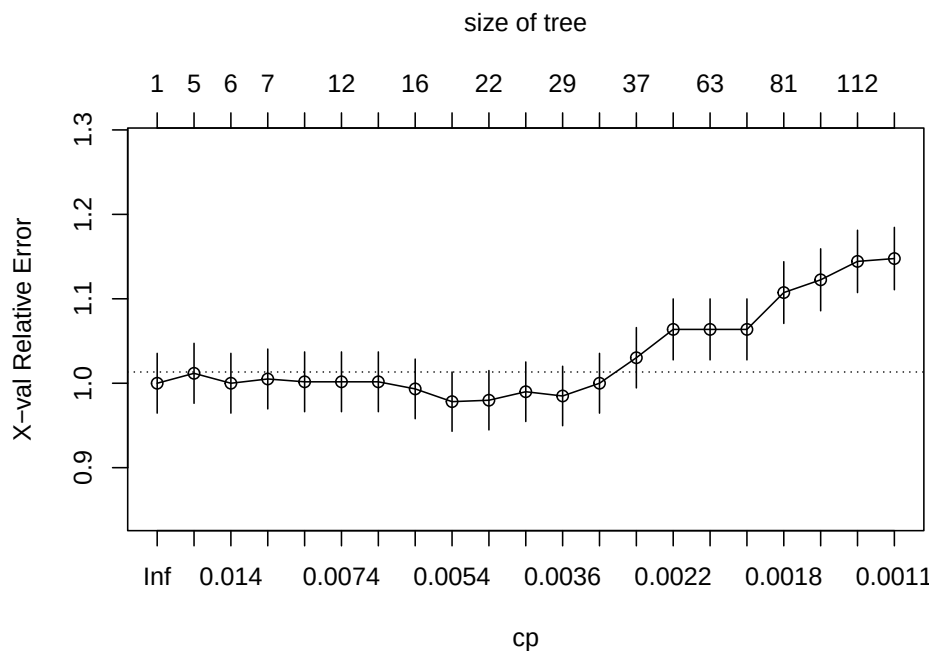
```



Looking at the error associated with different values of CP using the default options, it is not clear that we have the range that gives us the lowest error. We will use `part.control` to extend the tree and then prune it.

```
sbi.tree <- rpart(sbi.bin ~ fever_hours+age+sex+wcc+prevAB+pct+crp,
  data = sbi.data,
  method = "class",
  control = rpart.control(
    minsplit = 5, # min number of observ for a split
    minbucket = 5, # min nr of obs in terminal nodes
    cp=0.001)
  )

plotcp(sbi.tree)
```

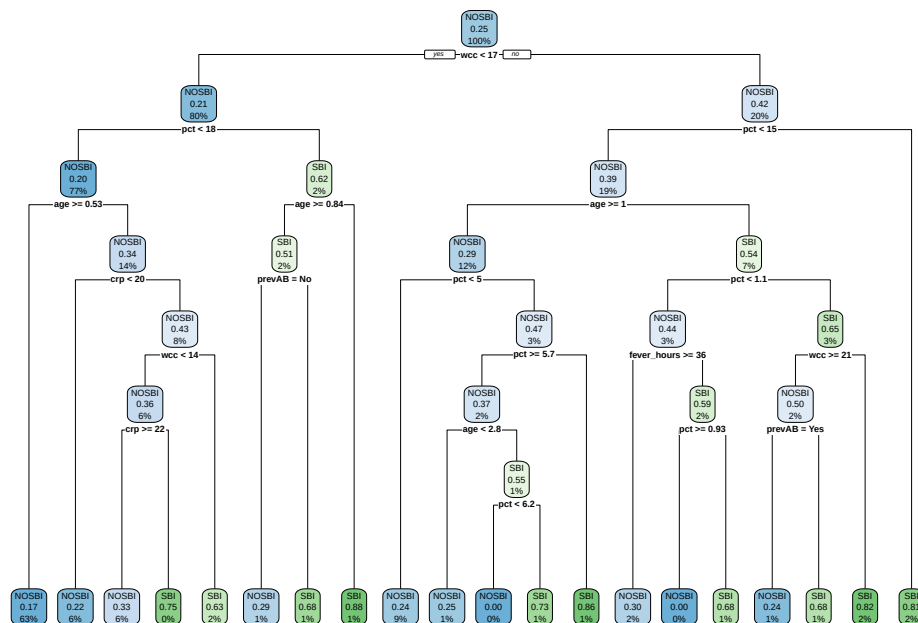


```
printcp(sbi.tree)
```

```
##
## Classification tree:
## rpart(formula = sbi.bin ~ fever_hours + age + sex + wcc + prevAB +
##      pct + crp, data = sbi.data, method = "class", control = rpart.control(minsplit = 5,
##      minbucket = 5, cp = 0.001))
##
## Variables actually used in tree construction:
## [1] age          crp          fever_hours pct          prevAB      sex          wcc
##
## Root node error: 596/2348 = 0.25383
##
## n= 2348
##
##      CP nsplit rel error  xerror    xstd
## 1 0.0201342    0  1.00000 1.00000 0.035383
## 2 0.0167785    4  0.91611 1.01174 0.035519
## 3 0.0117450    5  0.89933 1.00000 0.035383
## 4 0.0083893    6  0.88758 1.00503 0.035442
## 5 0.0075503    9  0.86242 1.00168 0.035403
## 6 0.0072707   11  0.84732 1.00168 0.035403
## 7 0.0067114   14  0.82550 1.00168 0.035403
## 8 0.0058725   15  0.81879 0.99329 0.035304
## 9 0.0050336   19  0.79530 0.97819 0.035125
```

If you have the same `set.seed(1999)` you should get `cp = 0.0050336` as the one with lowest error. We can now prune the tree accordingly.

```
sbi.prune.tree <- prune(sbi.tree, cp = 0.0050336)
rpart.plot(sbi.prune.tree)
```



TRY IT YOURSELF:

- 1) Use the `caret` package to fit the tree above.

See the solution code

```
trctrl <- trainControl(method = "repeatedcv",
                        number = 5,
                        repeats = 10,
```



```

classProbs = TRUE,
summaryFunction = twoClassSummary)

tree.sbi <- train(sbi.bin ~ fever_hours+age+sex+wcc+prevAB+pct+crp,
  data = sbi.data,
  method = "rpart",
  trControl = trctrl,
  tuneGrid=expand.grid(cp=seq(.001, .1, .001)))

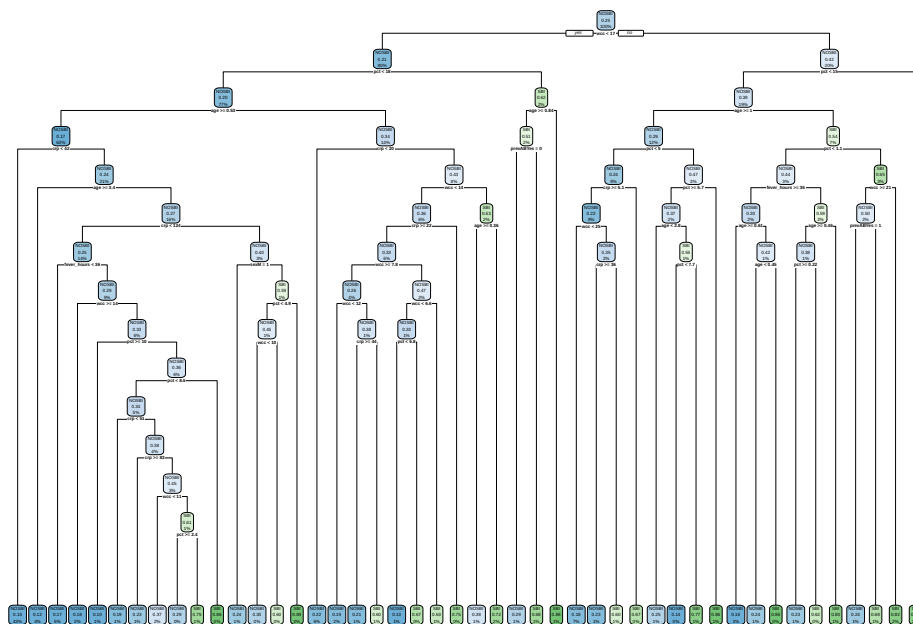
```

```
## Warning in train.default(x, y, weights = w, ...): The metric "Accuracy" was not in the result
## instead.
```

```

tree.sbi
tree.sbi$finalModel
rpart.plot(tree.sbi$finalModel)

```



- 2) Compare the confusion matrix of the trees fitted with `rpart()` function and the `caret` package.

```

Pred1 <- predict(sbi.prune.tree, sbi.data, type = "class" )
Pred2 <- predict(tree.sbi, sbi.data)

table(sbi.data$sbi.bin, Pred1)
table(sbi.data$sbi.bin, Pred2)

library(psych)
cohen.kappa(table(sbi.data$sbi.bin, Pred1))

```

```
cohen.kappa(table(sbi.data$sbi.bin, Pred2))
```

1.4 Exercises

Solve the following exercises:

- 1) The dataset SA_heart.csv contains on coronary heart disease status (variable *chd*) and several risk factors including the cumulative tobacco consumption *tobacco*, systolic *sbp*, and *age*
 - a) Fit a classification tree for *chd* using *tobacco*, *sbp* and *age*
 - b) Find the AUC ROC and confusion matrix for the tree
 - c) What is the predicted probability of coronary heart disease for someone with no tobacco consumption, *sbp*=132 and 45 years old?
- 2) The dataset fev.csv contains the measurements of forced expiratory volume (FEV) tests, evaluating the pulmonary capacity in 654 children and young adults.
 - a) Fit a regression tree to predict *fev* using *age*, *height* and *sex*.
 - b) Compare the MSE of the tree with a GAM model for *fev* with *sex* and smoothing splines for *height* and *age*.

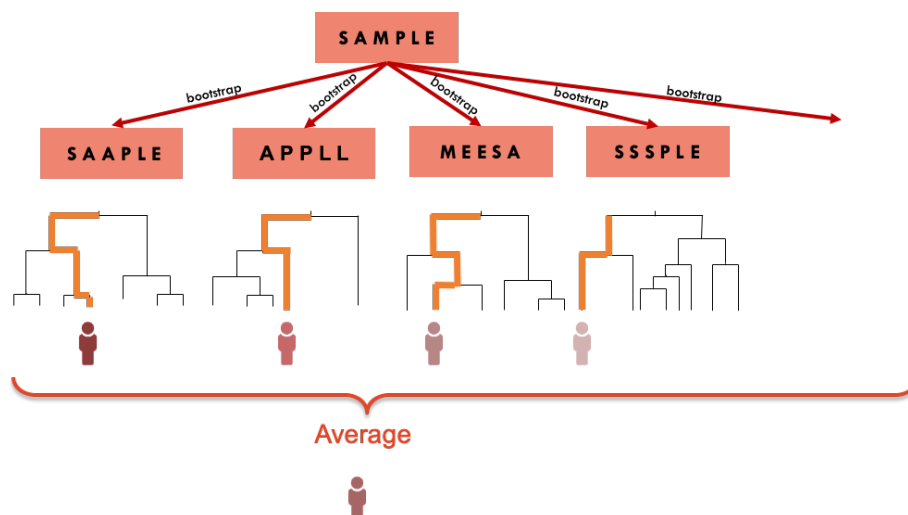
Chapter 2

Bagging

2.1 Introduction

Trees are quite flexible leading to non biased prediction, and intuitive to interpret. However, they tend to have a high variance, meaning that a small change in the data may result in a different tree.

Bagging is an approach that tries to “stabilise” the predictions from trees. Rather than fitting a single tree to the data, we will bootstrap the data and for each bootstrap sample, we will fit a tree. The final prediction is an average of all the predictions of the trees, if the outcome is continuous, or the class that was most common among the predictions, if the outcome is categorical



The downside of this approach is that we lose the interpretability of a single tree.

The result is a prediction from what in machine learning is sometimes referred as a *black box* method.

We can however use the multiple trees to create a metric of variable importance. The idea is that a variable that appears early in many trees is “more” important than a predictor that consistently appears further down in the trees.

2.2 Readings

Read the following chapters of *An introduction to statistical learning*:

- 8.2.1 Bagging

2.3 Practice session

Task 1 - Use bagging to build a classification model

The SBI.csv dataset contains the information of more than 2300 children that attended the emergency services with fever and were tested for serious bacterial infection. The variable `sbi` has 4 categories: Not Applicable(no infection) / UTI / Pneum / Bact

Create a new variable `sbi.bin` that identifies if a child was diagnosed or not with serious bacterial infection.

```
sbi.data <- read.csv("https://www.dropbox.com/s/wg32uj43fsy9yvd/SBI.csv?dl=1")
summary(sbi.data)
```

##	X	id	fever_hours	age	sex
##	Min. : 1.0	Min. : 495	Min. : 0.00	Min. : 0.010	Length : 2348
##	1st Qu.: 587.8	1st Qu.: 133039	1st Qu.: 24.00	1st Qu.: 0.760	N.unique : 2
##	Median : 1174.5	Median : 160016	Median : 48.00	Median : 1.525	N.blank : 0
##	Mean : 1174.5	Mean : 153698	Mean : 80.06	Mean : 1.836	Min.nchar: 1
##	3rd Qu.: 1761.2	3rd Qu.: 196030	3rd Qu.: 78.00	3rd Qu.: 2.752	Max.nchar: 1
##	Max. : 2348.0	Max. : 229986	Max. : 3360.00	Max. : 4.990	
##	sbi	pct	crp		
##	Length : 2348	Min. : 8.647e-03	Min. : 0.00		
##	N.unique : 4	1st Qu.: 1.600e-01	1st Qu.: 11.83		
##	N.blank : 0	Median : 7.600e-01	Median : 30.97		
##	Min.nchar: 3	Mean : 3.744e+00	Mean : 48.41		
##	Max.nchar: 13	3rd Qu.: 4.620e+00	3rd Qu.: 66.20		
##		Max. : 1.565e+02	Max. : 429.90		

```
# Create a binary variable based on "sbi"
sbi.data$sbi.bin <- as.factor(ifelse(sbi.data$sbi == "NotApplicable", "NOSBI", "SBI"))
table(sbi.data$sbi, sbi.data$sbi.bin)

##
```

```
##           NOSBI  SBI
##   Bact           0   34
##   NotApplicable 1752   0
##   Pneu           0  251
##   UTI            0  311
```

```
#sbi.data.sub <- sbi.data[,c("sbi.bin", "fever_hours", "age", "sex", "wcc", "prevAB", "pct", "crp")]
```

Now, let's using bagging to predict if a child has serious bacterial infection with the predictors **fever_hours**, **wcc**, **age**, **prevAB**, **pct**, and **crp**.

However, we will leave 200 observations out to test the model. We will use the `bagging()` function from the `ipred` package and fit 100 trees.

```
library(caret)
library(ipred) #includes the bagging function
library(rpart)
library(psych) # for the kappa statistics
set.seed(1999)

sbi.bag <- bagging(sbi.bin ~ fever_hours+age+sex+wcc+prevAB+pct+crp,
                  data = sbi.data[-c(500:700),], #not using rows 500 to 700
                  nbagg=100)
```

We will also fit one single tree to use as a comparison.

```
sbi.tree <- rpart(sbi.bin ~ fever_hours+age+sex+wcc+prevAB+pct+crp,
                 data = sbi.data[-c(500:700),], method="class",
                 control = rpart.control(cp=.001))
printcp(sbi.tree)
```

```
##
## Classification tree:
## rpart(formula = sbi.bin ~ fever_hours + age + sex + wcc + prevAB +
##       pct + crp, data = sbi.data[-c(500:700), ], method = "class",
##       control = rpart.control(cp = 0.001))
##
## Variables actually used in tree construction:
## [1] age          crp          fever_hours pct          prevAB       sex          wcc
##
## Root node error: 499/2147 = 0.23242
##
## n= 2147
##
##      CP nsplit rel error  xerror    xstd
## 1  0.021042      0  1.00000  1.00000  0.039220
## 2  0.018036      2  0.95792  0.98998  0.039083
## 3  0.015030      3  0.93988  0.98597  0.039027
```

```
## 4 0.010020      5 0.90982 0.98998 0.039083
## 5 0.008016      7 0.88978 0.96593 0.038745
## 6 0.007014     10 0.86573 0.97595 0.038887
## 7 0.006012     14 0.83166 0.98196 0.038971
## 8 0.004509     17 0.81363 0.98597 0.039027
## 9 0.004008     21 0.79559 0.99198 0.039110
## 10 0.003340    22 0.79158 0.99198 0.039110
## 11 0.003006    27 0.77355 1.01804 0.039464
## 12 0.002672    31 0.76152 1.02004 0.039491
## 13 0.002004    39 0.73747 1.03006 0.039624
## 14 0.001002    54 0.70741 1.10421 0.040558
## 15 0.001000    56 0.70541 1.11623 0.040701
```

```
sbi.tree <- prune(sbi.tree, cp=0.008016)
```

We can now compute the confusion matrix for both approaches, using the 200 cases left out from the model fitting.

```
sbi.test.data <- sbi.data[c(500:700),]
```

```
#confusion matrix for 1 tree
```

```
table(sbi.test.data$sbi.bin,
      predict(sbi.tree, sbi.test.data, type="class"))
```

```
##
##           NOSBI SBI
## NOSBI      94  10
## SBI        88   9
```

```
cohen.kappa(table(sbi.test.data$sbi.bin,
                  predict(sbi.tree, sbi.test.data, type="class")))
```

```
## Call: cohen.kappa1(x = x, w = w, n.obs = n.obs, alpha = alpha, levels = levels,
##      w.exp = w.exp)
##
## Cohen Kappa and Weighted Kappa correlation coefficients and confidence boundaries
##           lower estimate upper
## unweighted kappa -0.087  -0.0035  0.08
## weighted kappa  -0.087  -0.0035  0.08
##
## Number of subjects = 201
```

```
#confusion matrix for bagged trees
```

```
table(sbi.test.data$sbi.bin,
      predict(sbi.bag, sbi.test.data, type="class"))
```

```
##
##           NOSBI SBI
## NOSBI      95   9
```

```
## SBI      78 19
```

```
cohen.kappa(table(sbi.test.data$sbi.bin,
  predict(sbi.bag, sbi.test.data, type="class")))
```

```
## Call: cohen.kappa1(x = x, w = w, n.obs = n.obs, alpha = alpha, levels = levels,
##      w.exp = w.exp)
```

```
##
```

```
## Cohen Kappa and Weighted Kappa correlation coefficients and confidence boundaries
```

```
##           lower estimate upper
```

```
## unweighted kappa -0.0047    0.091 0.19
```

```
## weighted kappa  -0.0047    0.091 0.19
```

```
##
```

```
## Number of subjects = 201
```

The *bagging* improved the prediction ability.

TRY IT YOURSELF:

- 1) Use the `caret` package to fit the model above with bagging. You can use the entire dataset and compute the cross-validated AUC-ROC and confusion matrix

See the solution code

```
set.seed(1777)
trctrl    <- trainControl(method = "repeatedcv",
                          number = 10,
                          classProbs = TRUE,
                          summaryFunction = twoClassSummary)

bag.sbi    <- train(sbi.bin ~ fever_hours+age+sex+wcc+prevAB+pct+crp,
                  data = sbi.data,
                  method = "treebag",
                  trControl = trctrl,
                  nbagg = 100,
                  metric="ROC")

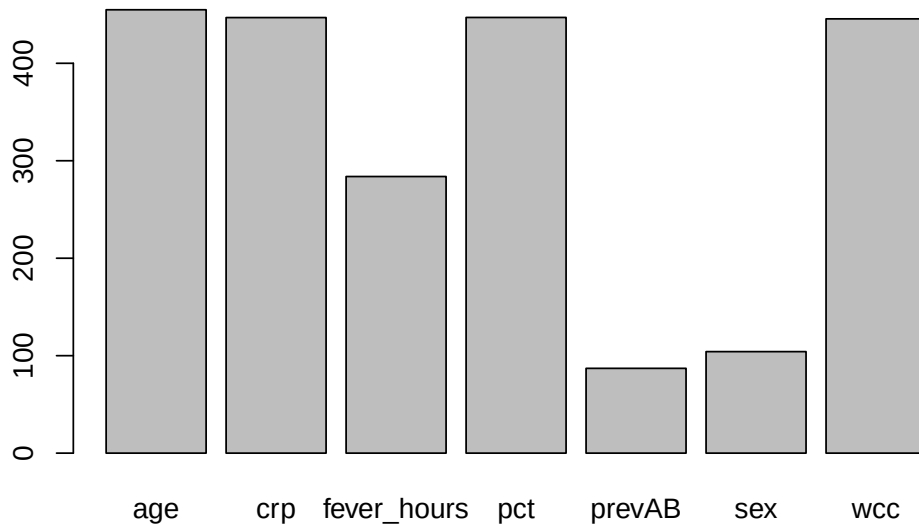
bag.sbi
confusionMatrix(bag.sbi)
```

Task 2 - Compute the variable importance

Let's compute the variable importance based on the `sbi.bag` bagging model. The function `varImp()` from the `caret` package, can be applied to the result of the `bagging()` function

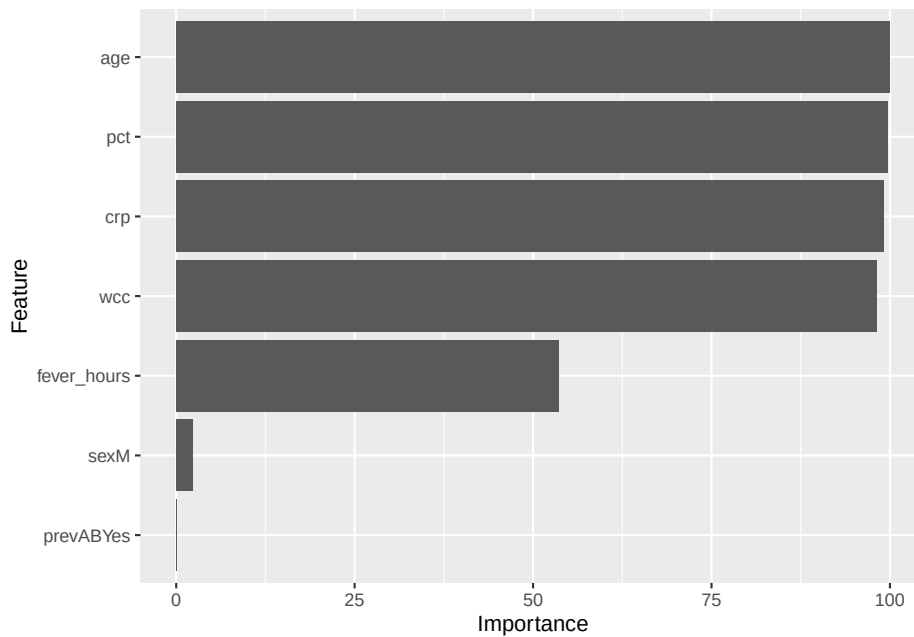
```
pred.imp <- varImp(sbi.bag)
pred.imp
```

```
#You can also plot the results  
barplot(pred.imp$Overall,  
        names.arg = row.names(pred.imp))
```



```
#If you use the caret package to fit the model, you can use  
#ggplot() to create an alternative plot
```

```
library(ggplot2)  
pred.imp.caret <- varImp(bag.sbi)  
ggplot(pred.imp.caret)
```

#This is a standard ggplot object which can be edited in the usual way

2.4 Exercises

Solve the following exercises:

- 1) The dataset `SA_heart.csv` contains on coronary heart disease status (variable *chd*) and several risk factors including the cumulative tobacco consumption *tobacco*, systolic *sbp*, and *age*
 - a) Build a predictive model by bagging 100 tree to classify *chd* using *tobacco*, *sbp* and *age*
 - b) Find the cross-validated AUC ROC and confusion matrix for the model above and compare them with ones obtained from logistic regression.
 - c) Which is the most important predictor?
 - d) What is the predicted probability of coronary heart disease for someone with no tobacco consumption, *sbp*=132 and 45 years old?
- 2) The dataset `fev.csv` contains the measurements of forced expiratory volume (FEV) tests, evaluating the pulmonary capacity in 654 children and young adults.
 - a) Fit a model based on bagging 200 tree to predict *fev* using *age*, *height* and *sex*.

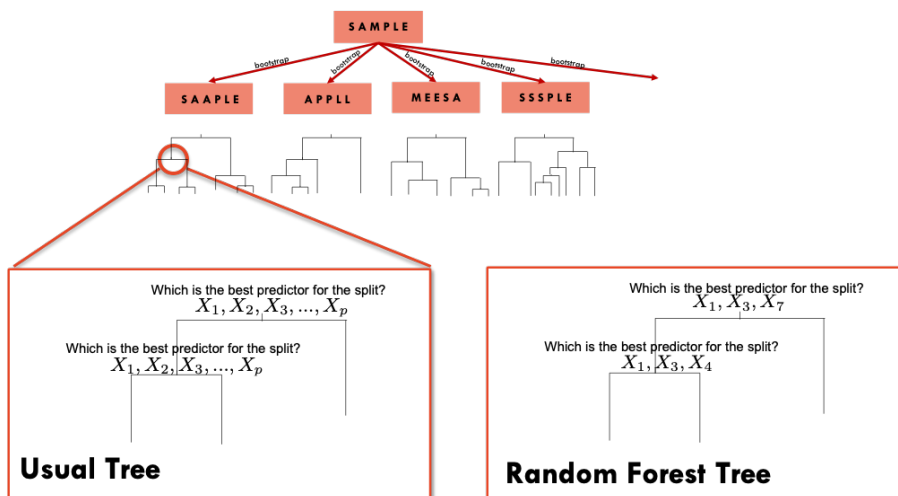
- b) Plot the variables importance.
- c) Compare the MSE of the model above with a GAM model for *fev* with *sex* and smoothing splines for *height* and *age*.

Chapter 3

Random Forests

3.1 Introduction

We have seen that **bagging** pools together the results of different trees based on bootstrap samples. These trees will tend to be highly correlated. **Random forests** use the same principle of bagged trees but with a difference in the construction of each tree. When growing a tree from one bootstrap sample, instead of evaluating all the predictors when deciding to split a node, **Random forests** only use a subset of the predictors randomly selected.



So, let's say that we have p predictors and rather than evaluating all the predictors at each knot, we select, for example 3 random predictors at each knot. Notice that the next knot will evaluate 3 different random predictors (unless, we get the same 3 by chance). This will uncorrelated the trees and produce better

predictions.

We can choose not only the number of trees but also the number of variables that will be considered at each node. A recommended option is to select $\sqrt{p}$ variables from the p predictors in a classification problem, and $p/3$ in a regression problem.

3.2 Readings

Read the following chapters of *An introduction to statistical learning*:

- 8.2.2 Random Forests

3.3 Practice session

Task 1 - Build a Random Forest model

The SBI.csv dataset contains the information of more than 2300 children that attended the emergency services with fever and were tested for serious bacterial infection. The variable **sbi** has 4 categories: Not Applicable(no infection) / UTI / Pneum / Bact

Create a new variable **sbi.bin** that identifies if a child was diagnosed or not with serious bacterial infection.

```
sbi.data <- read.csv("https://www.dropbox.com/s/wg32uj43fsy9yvd/SBI.csv?dl=1")
summary(sbi.data)
```

```
##           X           id      fever_hours      age      sex
## Min.      : 1.0    Min.      : 495    Min.      : 0.00    Min.      :0.010    Length :2348
## 1st Qu.: 587.8    1st Qu.:133039    1st Qu.: 24.00    1st Qu.:0.760    N.unique : 2
## Median :1174.5    Median :160016    Median : 48.00    Median :1.525    N.blank  : 0
## Mean    :1174.5    Mean    :153698    Mean     : 80.06    Mean     :1.836    Min.nchar: 1
## 3rd Qu.:1761.2    3rd Qu.:196030    3rd Qu.: 78.00    3rd Qu.:2.752    Max.nchar: 1
## Max.     :2348.0    Max.     :229986    Max.     :3360.00    Max.     :4.990
##           sbi           pct           crp
## Length      :2348    Min.      :8.647e-03    Min.      : 0.00
## N.unique     : 4    1st Qu.:1.600e-01    1st Qu.: 11.83
## N.blank      : 0    Median :7.600e-01    Median : 30.97
## Min.nchar:    3    Mean    :3.744e+00    Mean     : 48.41
## Max.nchar:   13    3rd Qu.:4.620e+00    3rd Qu.: 66.20
##           Max.      :1.565e+02    Max.      :429.90
```

```
# Create a binary variable based on "sbi"
```

```
sbi.data$sbi.bin <- as.factor(ifelse(sbi.data$sbi == "NotApplicable", "NOSBI", "SBI"))
table(sbi.data$sbi, sbi.data$sbi.bin)
```

```
##
```

```
##           NOSBI  SBI
##   Bact           0   34
##   NotApplicable 1752   0
##   Pneu           0  251
##   UTI            0  311
```

```
#sbi.data.sub <- sbi.data[,c("sbi.bin", "fever_hours", "age", "sex", "wcc", "prevAB", "pct", "crp")]
```

Now, let's using a Random Forest to predict if a child has serious bacterial infection with the predictors **fever_hours**, **wcc**, **age**, **prevAB**, **pct**, and **crp**.

However, we will leave 200 observations out to test the model. We will use the `randomForest()` function from the `randomForest` package and fit 100 trees with 2 predictors at each node ($\sqrt{7} = 2.6$)

```
library(caret)
library(randomForest) #includes the randomForest function
library(rpart)
library(psych) # for the kappa statistics
set.seed(1999)

rf.sbi <- randomForest(sbi.bin ~ fever_hours+age+sex+wcc+prevAB+pct+crp,
                       data = sbi.data[-c(500:700),], #not using rows 500 to 700
                       ntree=100, mtry=2)
```

We will also fit one single tree to use as a comparison.

```
sbi.tree <- rpart(sbi.bin ~ fever_hours+age+sex+wcc+prevAB+pct+crp,
                 data = sbi.data[-c(500:700),], method="class",
                 control = rpart.control(cp=.001))
printcp(sbi.tree)
```

```
##
## Classification tree:
## rpart(formula = sbi.bin ~ fever_hours + age + sex + wcc + prevAB +
##       pct + crp, data = sbi.data[-c(500:700), ], method = "class",
##       control = rpart.control(cp = 0.001))
##
## Variables actually used in tree construction:
## [1] age          crp          fever_hours pct          prevAB       sex          wcc
##
## Root node error: 499/2147 = 0.23242
##
## n= 2147
##
##           CP nsplit rel error xerror      xstd
## 1  0.021042     0   1.00000 1.0000 0.039220
## 2  0.018036     2   0.95792 1.0040 0.039275
```

```
## 3  0.015030      3  0.93988 1.0100 0.039357
## 4  0.010020      5  0.90982 1.0020 0.039248
## 5  0.008016      7  0.88978 1.0100 0.039357
## 6  0.007014     10  0.86573 1.0120 0.039384
## 7  0.006012     14  0.83166 1.0060 0.039302
## 8  0.004509     17  0.81363 1.0140 0.039411
## 9  0.004008     21  0.79559 1.0261 0.039571
## 10 0.003340     22  0.79158 1.0220 0.039518
## 11 0.003006     27  0.77355 1.0261 0.039571
## 12 0.002672     31  0.76152 1.0301 0.039624
## 13 0.002004     39  0.73747 1.0501 0.039885
## 14 0.001002     54  0.70741 1.1303 0.040866
## 15 0.001000     56  0.70541 1.1703 0.041321
```

```
sbi.tree <- prune(sbi.tree, cp=0.008016)
```

We can now compute the confusion matrix for both approaches, using the 200 cases left out from the model fitting.

```
sbi.test.data <- sbi.data[c(500:700),]
```

```
#confusion matrix for 1 tree
table(sbi.test.data$sbi.bin,
      predict(sbi.tree, sbi.test.data, type="class"))
```

```
##
##           NOSBI SBI
## NOSBI      94  10
## SBI        88   9
```

```
cohen.kappa(table(sbi.test.data$sbi.bin,
                  predict(sbi.tree, sbi.test.data, type="class")))
```

```
## Call: cohen.kappa1(x = x, w = w, n.obs = n.obs, alpha = alpha, levels = levels,
##      w.exp = w.exp)
##
## Cohen Kappa and Weighted Kappa correlation coefficients and confidence boundaries
##              lower estimate upper
## unweighted kappa -0.087  -0.0035  0.08
## weighted kappa   -0.087  -0.0035  0.08
##
## Number of subjects = 201
```

```
#confusion matrix for RF
table(sbi.test.data$sbi.bin,
      predict(rf.sbi, sbi.test.data, type="class"))
```

```
##
##           NOSBI SBI
```

```
## NOSBI    98    6
## SBI      84   13
```

```
cohen.kappa(table(sbi.test.data$sbi.bin,
  predict(rf.sbi, sbi.test.data, type="class")))
```

```
## Call: cohen.kappa1(x = x, w = w, n.obs = n.obs, alpha = alpha, levels = levels,
##      w.exp = w.exp)
##
## Cohen Kappa and Weighted Kappa correlation coefficients and confidence boundaries
##              lower estimate upper
## unweighted kappa -0.0054    0.078  0.16
## weighted kappa   -0.0054    0.078  0.16
##
## Number of subjects = 201
```

The *random forest* has a better prediction ability.

TRY IT YOURSELF:

- 1) Use the `caret` package to fit the model above. You can use the entire dataset and compute the cross-validated AUC-ROC and confusion matrix

See the solution code

```
set.seed(1777)
library(caret)
trctrl    <- trainControl(method = "repeatedcv",
                          number = 10,
                          repeats = 5,
                          classProbs = TRUE,
                          summaryFunction = twoClassSummary)

sbi.rf    <- train(sbi.bin ~ fever_hours+age+sex+wcc+prevAB+pct+crp,
                  data = sbi.data,
                  method = "rf",
                  trControl = trctrl,
                  ntree = 100,
                  tuneGrid = expand.grid(mtry = sqrt(7)),
                  metric="ROC")

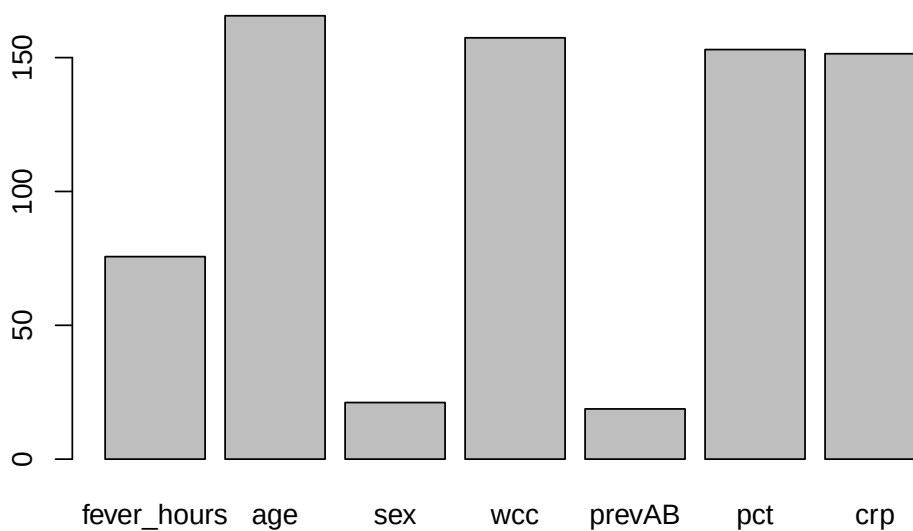
sbi.rf
confusionMatrix(sbi.rf)
```

Task 2 - Compute the variables importance

Let's compute the variable importance based on the `rf.sbi` random forest model.

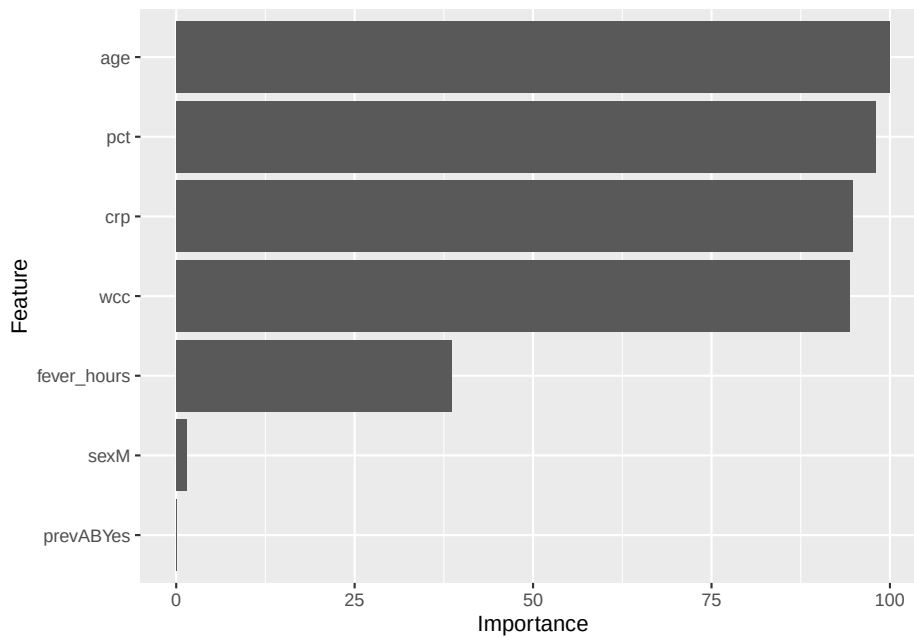
```
pred.imp <- varImp(rf.sbi)
pred.imp
```

```
#You can also plot the results  
barplot(pred.imp$Overall,  
        names.arg = row.names(pred.imp))
```



```
#If you use the caret package to fit the model, you can use  
#ggplot() to create an alternative plot
```

```
library(ggplot2)  
pred.imp.caret <- varImp(sbi.rf)  
ggplot(pred.imp.caret)
```

#This is a standard ggplot object which can be edited in the usual way

3.4 Exercises

Solve the following exercises:

- 1) The dataset `SA_heart.csv` contains on coronary heart disease status (variable `chd`) and several risk factors including the cumulative tobacco consumption `tobacco`, systolic `sbp`, and `age`
 - a) Build a predictive model using a random forest with 100 trees to classify `chd` using `tobacco`, `sbp` and `age`
 - b) Find the cross-validated AUC ROC and confusion matrix for the model above and compare them with ones obtained from logistic regression and bagging.
 - c) Which is the most important predictor?
 - d) What is the predicted probability of coronary heart disease for someone with no tobacco consumption, `sbp`=132 and 45 years old?
- 2) The dataset `fev.csv` contains the measurements of forced expiratory volume (FEV) tests, evaluating the pulmonary capacity in 654 children and young adults.
 - a) Fit a model based on random forest with 200 tree to predict `fev` using `age`, `height` and `sex`.

- b) Plot the variables importance.
- c) Compare the MSE of the model above with a GAM model for *fev* with *sex* and smoothing splines for *height* and *age*.